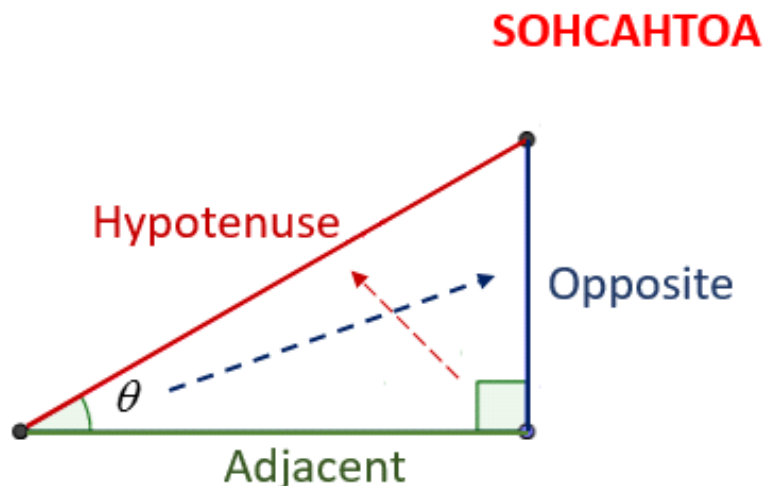


# Week 3 - Lesson 4 - Trigonometry Ratios

Wednesday, 8 April 2020 10:44 AM

Handy saying to help you know which ratio to use: **SOH CAH TOA**

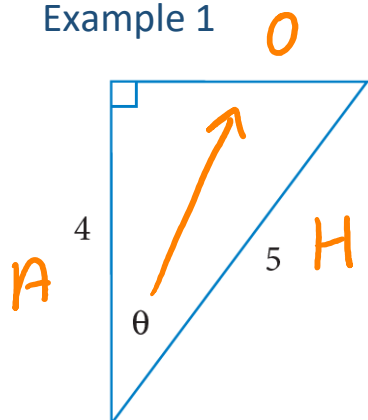


**SOH**  $\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$

**CAH**  $\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$

**TOA**  $\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$

Example 1



SOH CAH TOA

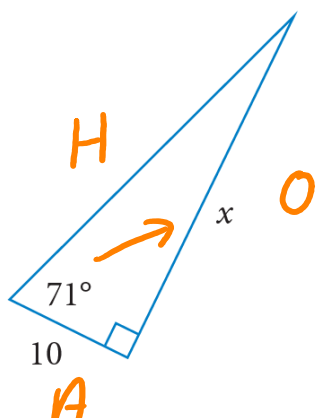
$\cos \theta = \frac{A}{H}$

$\cos \theta = \frac{4}{5}$

$\cos^{-1}(4 \div 5)$

$\theta = 36.87^\circ$

Example 2



SOH CAH TOA

$\tan \theta = \frac{O}{A}$

$\tan 71^\circ = \frac{x}{10}$

10  
A

10

$$x = 10 \times \tan 71^\circ$$

$$x = 29$$